Perfect — this is exactly how you should prepare for JNTU-GV exams. I'll give **crisp 2-mark answers**:

👉 **Definition + 1–2 key points (exam scoring style)**

🔷 UNIT–I: Introduction & Physical Layer

1. OSI Reference Model

The **OSI model** is a 7-layer framework used to understand network communication.
Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

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2. OSI vs TCP/IP (any 2)

- OSI has **7 layers**, TCP/IP has **4 layers**
- OSI is a **theoretical model**, TCP/IP is **practical implementation**

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3. LAN vs MAN vs WAN

- LAN: Small area (building)
- MAN: City-level network
- WAN: Large area (country/world)

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4. ARPANET

ARPANET was the first computer network developed by the U.S. Department of Defense, considered the **origin of the Internet**.

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5. Guided vs Unguided Media

- Guided: Uses physical cables (twisted pair, fiber)
- Unguided: Wireless transmission (radio waves, satellite)

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6. Multiplexing

Multiplexing is a technique to **combine multiple signals into one channel**.
Types: FDM, TDM, WDM.

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7. FDM vs TDM

- FDM: Divides frequency
- TDM: Divides time

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8. Statistical TDM

A TDM technique where time slots are allocated **based on demand**, improving efficiency.

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9. Digital Modulation

Conversion of digital data into analog signals for transmission.
Examples: ASK, FSK.

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10. Functions of Physical Layer

- Transmission of raw bits
- Defines voltage, data rate, connectors

1. What is bandwidth?

- Bandwidth is the **maximum data transmission capacity** of a channel, measured in bits per second (bps).
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2. Define attenuation.

- Attenuation is the **loss of signal strength** as it travels through a medium.
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3. What is noise in communication?

- Noise is any **unwanted disturbance** that affects signal quality.
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4. Define data rate.

- Data rate is the **number of bits transmitted per second**.
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5. What is a modem?

- A modem converts **digital signals to analog and vice versa**.
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6. Define propagation delay.

- Time taken by a signal to travel from sender to receiver.
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7. What is Nyquist Theorem?

- It defines the **maximum data rate for a noiseless channel**.
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8. What is Shannon Capacity?

- It gives the **maximum data rate in a noisy channel**.

🔷 UNIT–II: Data Link Layer

1. What is bit stuffing?

Adding extra bits in data to avoid confusion with frame boundaries.

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2. What is byte stuffing?

Inserting special bytes to distinguish data from control information.

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3. Define parity bit.

A parity bit is added to data to detect single-bit errors.

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4. What is checksum?

A method to detect errors by summing data values.

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5. What is MAC address?

A unique hardware address assigned to a network interface.

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6. What is collision in networks?

When two devices transmit simultaneously causing data corruption.

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7. What is CSMA/CA?

Carrier Sense Multiple Access with Collision Avoidance — used in wireless networks.

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8. Define throughput.

Actual rate of successful data transmission.

1. Framing

Framing is dividing data into **frames for transmission**.
Needed for synchronization and error detection.

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2. Error Detection & Correction

- Detection: Identifies errors (CRC)
- Correction: Fixes errors (Hamming code)

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3. Hamming Code

A method for **error detection and correction** using redundant bits.

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4. CRC

CRC is an error detection technique using **polynomial division** to detect errors.

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5. Flow Control

Controls data transmission speed between sender and receiver. Protocols: Stop-and-Wait, Sliding Window.

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6. Stop-and-Wait vs Sliding Window

- Stop-and-Wait: Sends one frame at a time
- Sliding Window: Sends multiple frames

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7. CSMA/CD

Carrier Sense Multiple Access with Collision Detection — detects and resolves collisions in Ethernet.

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8. ALOHA vs Slotted ALOHA

- ALOHA: Random transmission
- Slotted: Time slots reduce collisions

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9. VLAN

Virtual LAN logically divides a network for better management and security.

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10. Token Ring vs Token Bus

- Token Ring: Ring topology

- Token Bus: Bus topology

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11. Bridges

Devices that **connect two LANs** and filter traffic based on MAC addresses.

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12. IEEE 802.11 & 802.16

- 802.11: Wi-Fi
- 802.16: WiMAX

🔷 UNIT–III: Network Layer

1. What is hop count?

Number of routers a packet passes through.

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2. Define delay in networking.

Time taken for data to travel from source to destination.

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3. What is packet switching?

Data is divided into packets and sent independently.

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4. What is circuit switching?

A dedicated path is established before transmission.

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5. Define gateway.

A device that connects different networks using different protocols.

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6. What is ICMP?

Internet Control Message Protocol used for error reporting.

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7. What is NAT?

Network Address Translation converts private IP to public IP.

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8. What is TTL?

Time To Live limits the lifetime of a packet.

1. Routing

Routing is the process of selecting paths for data transmission.

- Static: Fixed
- Dynamic: Changes automatically

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2. Shortest Path Routing

Selects the path with **minimum cost/distance**.

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3. Flooding

Each packet is sent to **all possible paths** in the network.

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4. Distance Vector vs Link State

- Distance Vector: Uses neighbor info
- Link State: Uses full network topology

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5. Congestion

Network overload due to excess traffic, causing delay and packet loss.

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6. Choke Packets

Packets sent by routers to **slow down sender transmission rate**.

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7. Load Shedding

Dropping packets when network is overloaded.

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8. Jitter Control

Reduces variation in packet delay for smooth data delivery.

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9. Tunneling

Encapsulation of one protocol inside another for transmission.

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10. Fragmentation

Breaking large packets into smaller ones for transmission.

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11. IP Address

Unique identifier for a device in a network.

- IPv4: 32-bit
- IPv6: 128-bit

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12. Subnet & Masking

Subnet divides network into smaller parts.
Subnet mask identifies network and host portions.

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13. OSPF & BGP

- OSPF: Interior routing protocol
- BGP: Exterior routing protocol

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14. Mobile IP

Allows devices to **move between networks without changing IP**.

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15. Multicasting vs Broadcasting

- Multicasting: One-to-many (specific group)
- Broadcasting: One-to-all

🔷 UNIT–IV: Transport Layer

1. What is multiplexing in transport layer?

Combining multiple application data streams into one.

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2. What is demultiplexing?

Delivering received data to the correct application.

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3. Define port number.

A port number identifies a specific process or service.

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4. What is segmentation?

Breaking data into smaller segments for transmission.

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5. What is congestion control in TCP?

Mechanism to prevent network overload.

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6. What is slow start algorithm?

TCP increases transmission rate gradually at the start.

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7. What is piggybacking?

Combining ACK with data to improve efficiency.

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8. What is urgent data in TCP?

Data that needs immediate processing.

1. Functions of Transport Layer

- End-to-end communication
- Flow control
- Error control

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2. TCP vs UDP

- TCP: Reliable
- UDP: Fast but unreliable

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3. Socket

Combination of IP address + port number for communication.

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4. Connection Establishment & Termination

Establishment: Three-way handshake
Termination: Four-step process

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5. Three-Way Handshake

1. SYN
2. SYN-ACK
3. ACK

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6. Flow Control in TCP

Uses sliding window to control transmission rate.

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7. Crash Recovery

Recovery from failure using retransmission and checkpoints.

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8. RTP

Real-Time Transport Protocol used for audio/video streaming.

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9. Sliding Window in TCP

Allows sending multiple segments before receiving ACK.

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10. Adaptive Retransmission

Adjusts timeout dynamically based on network conditions.

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11. RPC

Remote Procedure Call allows calling functions on remote systems.

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12. TCP Segment Format

Contains fields: Source port, Destination port, Sequence number, ACK number, Flags, Window size.

🔷 UNIT–V: Application Layer

1. What is FTP?

File Transfer Protocol used to transfer files over network.

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2. What is Telnet?

Protocol used for remote login.

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3. What is HTTPS?

Secure version of HTTP using encryption.

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4. Define web browser.

Software used to access web pages (e.g., Chrome).

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5. What is caching?

Storing frequently accessed data for faster retrieval.

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6. What is proxy server?

Acts as an intermediary between client and server.

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7. What is cookie?

Small data stored on user's system by websites.

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8. What is firewall?

Security system that monitors and controls network traffic.

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1. WWW

World Wide Web is a system of interlinked web pages using HTTP.

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2. URL

Uniform Resource Locator — address of a resource.
Example: https://www.google.com

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3. HTTP

Protocol used for web communication.
Methods: GET, POST

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4. HTTP Request vs Response

- Request: Sent by client
- Response: Sent by server

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5. DNS

Domain Name System converts domain names into IP addresses.

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6. DNS Name Space

Hierarchical structure of domain names.

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7. Resource Records

DNS entries storing information like IP addresses (A record).

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8. SMTP

Protocol used for sending emails.

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9. POP3 vs IMAP

- POP3: Downloads emails
- IMAP: Syncs emails on server

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10. MIME

Multipurpose Internet Mail Extensions allow sending multimedia emails.

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11. User Agent

Software used to send/receive emails (e.g., Gmail).

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12. Email Format

Contains:

- Header (To, From, Subject)
- Body (message content)